Mid-Semester Exam — Solutions

Numerical Algorithms

Time: 1.5 hours | Total: 100 points

Notes: These are model solutions. Equivalent correct methods and equivalent final answers (within rounding) earn full credit.

Part A: True/False ($10 \times 1 = 10$ points)

1. **T**. Forward difference has truncation error $O(h)$.

2. **T**. Central difference has truncation error $O(h^2)$.

3. **T**. Composite trapezoidal rule is exact for polynomials up to degree 1.

4. **F**. Simpson's 1/3 rule requires an *even* number of subintervals.

5. **T**. In convex standard form, equality constraints must be affine.

6. **T**. $g(\lambda, \nu) = \inf_x L(x, \lambda, \nu)$ is concave as an infimum of affine functions in (λ, ν).

7. **F**. For a minimization primal, weak duality gives $d^* \leq p^*$ (dual value is a lower bound).

8. **T**. Complementary slackness: $\lambda_i^* f_i(x^*) = 0$ for each inequality constraint.

9. **F**. Experience replay is *not* required for tabular Q-learning to converge.

10. **T**. The Bellman optimality equation includes $\max_{a'} Q^*(s', a')$, motivating the Q-learning target.

Part B: Short Answer ($6 \times 5 = 30$ points)

11. Standard form and feasible set.

$$\begin{aligned} \text{minimize} \quad & (x-1)^2 + 2 \\ \text{subject to} \quad & 2x - 3 \geq 0, \\ & x \leq 4. \end{aligned}$$

(a) Standard form uses inequalities $f_i(x) \leq 0$:

$$f_0(x) = (x-1)^2 + 2, \qquad f_1(x) = 3 - 2x \leq 0, \qquad f_2(x) = x - 4 \leq 0.$$

(b) Feasible set: $2x - 3 \geq 0 \Rightarrow x \geq \frac{3}{2}$ and $x \leq 4$, hence

$$\{x : x \geq 3/2, \ x \leq 4\} = [3/2, \ 4].$$

12. Convexity check. $f(x) = -\log x$, domain $(0, \infty)$.

(a)

$$f'(x) = -\frac{1}{x}, \qquad f''(x) = \frac{1}{x^2}.$$

For all $x > 0$, $f''(x) = 1/x^2 > 0$, so f is convex on $(0, \infty)$.

(b) Domain: $\operatorname{dom} f = (0, \infty)$.

13. Conjugate and reflexivity (biconjugate).

(a) The Fenchel conjugate of $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is

$$f^*(y) = \sup_{x \in \operatorname{dom} f} \big(y^T x - f(x)\big).$$

(b) For $f(x) = \frac{1}{2}x^2$ (scalar x):

$$f^*(y) = \sup_{x \in \mathbb{R}} \left(yx - \frac{1}{2}x^2\right).$$

Maximize the concave quadratic in x: derivative $y - x = 0 \Rightarrow x = y$. Substituting:

$$f^*(y) = y \cdot y - \frac{1}{2}y^2 = \frac{1}{2}y^2.$$

(c) Since f is closed, proper, and convex, $f^{**} = f$. Indeed,

$$f^{**}(x) = \frac{1}{2}x^2 = f(x).$$

14. Weak duality and complementary slackness.

(a) (Weak duality, minimization primal.) For $\lambda \geq 0$ and any ν,

$$g(\lambda, \nu) = \inf_{x \in D} L(x, \lambda, \nu) \leq L(\tilde{x}, \lambda, \nu)$$

for any feasible $\tilde{x}$. If $\tilde{x}$ is feasible, then $f_i(\tilde{x}) \leq 0$ and $h_j(\tilde{x}) = 0$, hence

$$L(\tilde{x}, \lambda, \nu) = f_0(\tilde{x}) + \sum_i \lambda_i f_i(\tilde{x}) + \sum_j \nu_j h_j(\tilde{x}) \leq f_0(\tilde{x}).$$

So $g(\lambda, \nu) \leq f_0(\tilde{x})$ for all feasible $\tilde{x}$, hence $g(\lambda, \nu) \leq p^*$. Taking supremum over dual-feasible (λ, ν) gives $d^* \leq p^*$.

(b) Complementary slackness (inequalities): if (x^*, λ^*, ν^*) is primal-dual optimal with zero duality gap, then

$$\lambda_i^* f_i(x^*) = 0, \qquad i = 1, \ldots, m.$$

15. RL fundamentals and Bellman equations.

(a) The (discounted) state-value function under policy π:

$$V^\pi(s) = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \;\middle|\; S_t = s\right].$$

The action-value function:

$$Q^\pi(s,a) = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \;\middle|\; S_t = s,\ A_t = a\right].$$

(b) Bellman expectation equation for V^π:

$$V^\pi(s) = \mathbb{E}_\pi\big[R_{t+1} + \gamma V^\pi(S_{t+1}) \mid S_t = s\big] = \sum_a \pi(a|s) \sum_{s'} P(s'|s,a)\Big(r(s,a,s') + \gamma V^\pi(s')\Big).$$

(c) Q-learning uses a Bellman equation because Q^* is the fixed point of the Bellman *optimality* operator, and the update is a stochastic approximation to that fixed-point equation.

16. State/action representations in RL.

(a) **Chess.** A reasonable state: full board configuration (piece type and location on 8×8), side-to-move, castling rights, en-passant availability, and move counters (for 50-move rule/threefold bookkeeping if desired). A reasonable action set: all legal moves from the position (including captures, promotions, castling, en-passant).

(b) **Cart-pole.** State: $(x, \dot{x}, \theta, \dot{\theta})$ (cart position/velocity, pole angle/angular velocity). Action set: typically $\{\text{push left}, \text{push right}\}$ (discrete force).

(c) **Tetris/brick game.** State: occupancy grid (stack heights / full binary grid), current falling piece (type, position, orientation), and optionally next piece/hold piece. Actions: move left/right, rotate, soft drop, hard drop, (optional hold).

(d) **Atari from pixels.** States are commonly represented as a stack of recent image frames (e.g., several grayscale frames) to encode motion. A CNN maps pixels to feature representations (spatial/semantic features) used by the value network (or Q-network) for decision making.

Part C: Long Answer ($3 \times 20 = 60$ points)

17. Numerical differentiation, numerical integration, and Newton's method. Let $f(x) = e^x$ and $h = 0.1$.

(a) Approximate $f'(0.2)$.

Compute:

$$f(0.1) = e^{0.1} \approx 1.1051709181, \quad f(0.2) = e^{0.2} \approx 1.2214027582, \quad f(0.3) = e^{0.3} \approx 1.3498588076.$$

(i) Forward difference:

$$f'(0.2) \approx \frac{f(0.3) - f(0.2)}{0.1} = \frac{1.3498588076 - 1.2214027582}{0.1} \approx 1.2845604942.$$

(ii) Central difference:

$$f'(0.2) \approx \frac{f(0.3) - f(0.1)}{0.2} = \frac{1.3498588076 - 1.1051709181}{0.2} \approx 1.2234394475.$$

(b) Approximate $f''(0.2)$ and compute absolute error.
Second-order central difference:

$$f''(0.2) \approx \frac{f(0.3) - 2f(0.2) + f(0.1)}{h^2} = \frac{1.3498588076 - 2(1.2214027582) + 1.1051709181}{0.01} \approx 1.22242$$

Exact:

$$f''(0.2) = e^{0.2} \approx 1.2214027582.$$

Absolute error:

$$|1.2224209331 - 1.2214027582| \approx 0.0010181750.$$

(c) Composite Simpson's rule with $n = 4$ subintervals on $[0, 0.4]$.
Here $h = (0.4 - 0)/4 = 0.1$ and

$$f(0) = 1, \quad f(0.1) \approx 1.1051709181, \quad f(0.2) \approx 1.2214027582,$$

$$f(0.3) \approx 1.3498588076, \quad f(0.4) = e^{0.4} \approx 1.4918246976.$$

Composite Simpson:

$$I \approx \frac{h}{3}\left[f(0) + f(0.4) + 4\big(f(0.1) + f(0.3)\big) + 2f(0.2)\right].$$

Numerically:

$$I \approx \frac{0.1}{3}\left[1 + 1.4918246976 + 4(1.1051709181 + 1.3498588076) + 2(1.2214027582)\right] \approx 0.4918249706.$$

Exact:

$$I_{\text{exact}} = \int_0^{0.4} e^x \, dx = e^{0.4} - 1 \approx 1.4918246976 - 1 = 0.4918246976.$$

Absolute error:

$$|I - I_{\text{exact}}| \approx |0.4918249706 - 0.4918246976| \approx 2.73 \times 10^{-7}.$$

(d) Newton's method for minimizing $g(x) = e^x - 2x$.

Compute derivatives:

$$g'(x) = e^x - 2, \qquad g''(x) = e^x.$$

Newton update:

$$x_{k+1} = x_k - \frac{g'(x_k)}{g''(x_k)} = x_k - \frac{e^{x_k} - 2}{e^{x_k}} = x_k - 1 + 2e^{-x_k}.$$

Start $x_0 = 0$:

$$x_1 = 0 - 1 + 2e^0 = 1.$$
$$x_2 = 1 - 1 + 2e^{-1} = 2e^{-1} \approx 0.7357588823.$$
$$x_3 = x_2 - 1 + 2e^{-x_2} \approx 0.7357588823 - 1 + 2e^{-0.7357588823} \approx 0.6940422999.$$

The minimizer satisfies the first-order condition $g'(x) = 0 \Rightarrow e^x = 2 \Rightarrow x^* = \ln 2 \approx 0.6931471806$, so $x_k \to \ln 2$.

18. Convex optimization, duality, and KKT.

$$\begin{aligned} \text{minimize} \quad & \frac{1}{2}(x_1^2 + x_2^2) \\ \text{subject to} \quad & x_1 + x_2 \ge 1, \\ & x_1 \ge 0, \;\; x_2 \ge 0. \end{aligned}$$

(a) Standard form $f_i(x) \le 0$:

Objective:

$$f_0(x) = \frac{1}{2}(x_1^2 + x_2^2).$$

Constraints:

$$x_1 + x_2 \ge 1 \iff 1 - x_1 - x_2 \le 0 \quad \Rightarrow \quad f_1(x) = 1 - x_1 - x_2,$$

$$x_1 \ge 0 \iff -x_1 \le 0 \quad \Rightarrow \quad f_2(x) = -x_1,$$
$$x_2 \ge 0 \iff -x_2 \le 0 \quad \Rightarrow \quad f_3(x) = -x_2.$$

(b) Convexity and Slater.

$f_0(x) = \frac{1}{2}\|x\|_2^2$ is convex (quadratic with positive semidefinite Hessian). All constraints are affine, hence convex.

Slater's condition (for inequality-only convex problems): there exists x such that all inequalities are *strict*:

$$1 - x_1 - x_2 < 0, \quad -x_1 < 0, \quad -x_2 < 0$$

i.e., $x_1 > 0$, $x_2 > 0$, $x_1 + x_2 > 1$. For example, $x = (1, 1)$ is strictly feasible, so Slater holds, hence strong duality holds.

(c) Lagrangian and dual function.
Let multipliers $\lambda_1, \lambda_2, \lambda_3 \geq 0$ correspond to f_1, f_2, f_3.
Lagrangian:

$$L(x, \lambda) = \frac{1}{2}(x_1^2 + x_2^2) + \lambda_1(1 - x_1 - x_2) + \lambda_2(-x_1) + \lambda_3(-x_2).$$

To minimize over x, set partial derivatives to zero:

$$\frac{\partial L}{\partial x_1} = x_1 - \lambda_1 - \lambda_2 = 0 \Rightarrow x_1 = \lambda_1 + \lambda_2,$$

$$\frac{\partial L}{\partial x_2} = x_2 - \lambda_1 - \lambda_3 = 0 \Rightarrow x_2 = \lambda_1 + \lambda_3.$$

Substitute into L to get $g(\lambda) = \inf_x L(x, \lambda)$:

$$g(\lambda) = \lambda_1 - \frac{1}{2}\Big((\lambda_1 + \lambda_2)^2 + (\lambda_1 + \lambda_3)^2\Big), \qquad \lambda \geq 0.$$

(d) Dual problem and solution.
Dual:

$$\begin{array}{ll} \text{maximize} & g(\lambda) = \lambda_1 - \frac{1}{2}\Big((\lambda_1 + \lambda_2)^2 + (\lambda_1 + \lambda_3)^2\Big) \\ \text{subject to} & \lambda_1, \lambda_2, \lambda_3 \geq 0. \end{array}$$

Note $(\lambda_1 + \lambda_2)^2 \geq \lambda_1^2$ and $(\lambda_1 + \lambda_3)^2 \geq \lambda_1^2$, so

$$g(\lambda) \leq \lambda_1 - \frac{1}{2}(\lambda_1^2 + \lambda_1^2) = \lambda_1 - \lambda_1^2.$$

Maximizing $\lambda_1 - \lambda_1^2$ over $\lambda_1 \geq 0$ gives

$$\frac{d}{d\lambda_1}(\lambda_1 - \lambda_1^2) = 1 - 2\lambda_1 = 0 \Rightarrow \lambda_1^* = \frac{1}{2},$$

and the maximum value is

$$d^* = \lambda_1^* - (\lambda_1^*)^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}.$$

This bound is achievable by taking $\lambda_2^* = \lambda_3^* = 0$, hence

$$\lambda^* = \left(\frac{1}{2}, 0, 0\right), \qquad d^* = \frac{1}{4}.$$

(e) Recover primal optimum and verify complementary slackness.
From the minimizing $x(\lambda)$:

$$x_1^* = \lambda_1^* + \lambda_2^* = \frac{1}{2}, \qquad x_2^* = \lambda_1^* + \lambda_3^* = \frac{1}{2}.$$

Check feasibility: $x_1^* + x_2^* = 1$, $x_1^* \geq 0$, $x_2^* \geq 0$.
Complementary slackness:

$$\lambda_1^*(1 - x_1^* - x_2^*) = \frac{1}{2}(1-1) = 0,$$

$$\lambda_2^*(-x_1^*) = 0 \cdot (-1/2) = 0, \qquad \lambda_3^*(-x_2^*) = 0 \cdot (-1/2) = 0.$$

Also $f_0(x^*) = \frac{1}{2}\left((1/2)^2 + (1/2)^2\right) = \frac{1}{4} = d^*$, consistent with strong duality.

19. Q-learning on a grid world (no experience replay).

States: s_1 (start), s_2, s_3, and s_T terminal. Actions: $\{R, D\}$. Rewards: entering s_T gives $+5$ and ends; otherwise -1. $\gamma = 0.9$, $\alpha = 0.5$, and $Q_0(s,a) = 0$.

(a) Tabular Q-learning update:

$$Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) + \alpha \left(r_{t+1} + \gamma \max_{a'} Q_t(s_{t+1}, a') - Q_t(s_t, a_t) \right).$$

Meanings (one short sentence each): $Q_t(s_t, a_t)$ is the current estimate; $r_{t+1} + \gamma \max_{a'} Q_t(s_{t+1}, a')$ is the one-step bootstrap target; the bracket is the TD error; α is the step size.

(b) Perform updates in chronological order.

Episode 1, transition 1: $s_1 \xrightarrow{R,-1} s_2$.
Initially $Q(s_1, R) = 0$ and $\max_{a'} Q(s_2, a') = \max(0, 0) = 0$.

$$Q(s_1, R) \leftarrow 0 + 0.5\,(-1 + 0.9 \cdot 0 - 0) = -0.5.$$

After this update:

$$Q(s_1, R) = -0.5, \qquad Q(s_2, D) = 0.$$

Episode 1, transition 2: $s_2 \xrightarrow{D,+5} s_T$ (terminal).
Take $\max_{a'} Q(s_T, a') = 0$ (terminal bootstrap).

$$Q(s_2, D) \leftarrow 0 + 0.5\,(5 + 0.9 \cdot 0 - 0) = 2.5.$$

After this update:

$$Q(s_1, R) = -0.5, \qquad Q(s_2, D) = 2.5.$$

Episode 2, transition 1: $s_1 \xrightarrow{R,-1} s_2$.
Now $\max_{a'} Q(s_2, a') = \max(Q(s_2, R), Q(s_2, D)) = \max(0, 2.5) = 2.5$.

$$Q(s_1, R) \leftarrow -0.5 + 0.5\,(-1 + 0.9 \cdot 2.5 - (-0.5)).$$

Compute the target: $-1 + 0.9 \cdot 2.5 = -1 + 2.25 = 1.25$. So TD error $= 1.25 - (-0.5) = 1.75$ and

$$Q(s_1, R) \leftarrow -0.5 + 0.5(1.75) = -0.5 + 0.875 = 0.375.$$

After this update:

$$Q(s_1, R) = 0.375, \qquad Q(s_2, D) = 2.5.$$

Episode 2, transition 2: $s_2 \xrightarrow{D,+5} s_T$.

$$Q(s_2, D) \leftarrow 2.5 + 0.5\,(5 - 2.5) = 2.5 + 1.25 = 3.75.$$

After this update:

$$Q(s_1, R) = 0.375, \qquad Q(s_2, D) = 3.75.$$

(c) Greedy actions and values after Episode 2.

At s_1: $Q(s_1, R) = 0.375$ and $Q(s_1, D) = 0$, so greedy action is R and

$$V(s_1) = \max_a Q(s_1, a) = 0.375.$$

At s_2: $Q(s_2, D) = 3.75$ and $Q(s_2, R) = 0$, so greedy action is D and

$$V(s_2) = \max_a Q(s_2, a) = 3.75.$$

(d) Experience replay.

Experience replay stores transitions (s, a, r, s') in a buffer and trains by sampling (often uniformly) from the buffer. One reason it can help in deep Q-learning: it breaks temporal correlations between consecutive samples and improves sample efficiency by reusing past data, which stabilizes learning with function approximation.

(e) Bellman optimality equation and convergence of tabular Q-learning.

Bellman optimality equation for Q^*:

$$Q^*(s, a) = \mathbb{E}\Big[R_{t+1} + \gamma \max_{a'} Q^*(S_{t+1}, a') \;\Big|\; S_t = s,\ A_t = a\Big].$$

Two standard conditions for convergence (tabular setting):

- **Sufficient exploration:** every state-action pair (s, a) is visited infinitely often (e.g., persistent exploration such as GLIE).
- **Learning-rate (Robbins–Monro):** for each (s, a), $\alpha_t(s, a) \in (0, 1]$ with $\sum_t \alpha_t(s, a) = \infty$ and $\sum_t \alpha_t(s, a)^2 < \infty$ (and bounded rewards, $\gamma \in [0, 1)$).

Why the Bellman optimality operator matters: it is a γ-contraction in the sup-norm, so it has a unique fixed point Q^*; Q-learning is a stochastic approximation method that converges to this fixed point under the above conditions.

End of Solutions